1. $z = w^T x + b$

$$= \begin{bmatrix} 1 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} - 2$$

$$= \begin{bmatrix} 2 & -3 \end{bmatrix} - 2 = -3$$

2. $z = -0.5 \cdot 4 + 1 = -1$

$$\hat{y} = \sigma(-1) = \frac{1}{1 + e^{-(-1)}} \approx 0.269$$

3. $$z_1 = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1-2 \\ 2+0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$$h = \mathrm{ReLU}(z_1) = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$y = \begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \end{bmatrix} + 0$$

$$= 0 - 3 + 0 = -3$$

4. $L = \frac{1}{2}(-1-2)^2 = \frac{9}{2}$

5. $L = \frac{1}{2}(\hat{y}-y)^2 \quad u = \hat{y}-y \quad \partial u = 1$

$$\frac{\partial L}{\partial \hat{y}} = (\hat{y}-y)\cdot 1 = (\hat{y}-y)$$

6. $L = \frac{1}{2}(w^T x + b - y)^2$

$$\hat{y} = \begin{bmatrix} 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -2 \end{bmatrix} + 1 = -1$$

$$\frac{\partial L}{\partial w_1} = (\hat{y}-y)\cdot x_1 = -1\cdot 1 = -1$$

$$\frac{\partial L}{\partial w_2} = (\hat{y}-y)\cdot x_2 = -1\cdot -2 = 2$$

$$\frac{\partial L}{\partial b} = (\hat{y}-y) = -1$$

7. $w_{1new} \leftarrow 0 - 0.2(-1) = 0.2$

$w_{2new} \leftarrow 1 - 0.2(2) = 0.6$

$b_{new} \leftarrow 1 - 0.2(-1) = 1.2$

8. $z = 2 \cdot -1 - 1 = -3 \quad a = ReLU(-3) = 0$

$$\frac{\partial L}{\partial w} = (a - y) \cdot \frac{\partial ReLU(z)}{\partial z} \cdot x = (0 - 3) \cdot 0 \cdot -1 = 0$$

$$\frac{\partial L}{\partial b} = (a - y) \cdot \frac{\partial ReLU(z)}{\partial b} = (0 - 3) \cdot 0 = 0$$

9. $\hat{y} = (0 \cdot 2) + 0 = 0$

$$\frac{\partial L}{\partial w} = (\hat{y} - y) \cdot x = (0 - 1) \cdot 2 = -2$$

$$\frac{\partial L}{\partial b} = (\hat{y} - y) = (0 - 1) = -1$$

$w_{1new} \leftarrow 0 - 0.1 \cdot -2) = 0.2$

$b_{1new} \leftarrow 0 - 0.1 \cdot -1 = 0.1$

$\hat{y} = 0.2 \cdot 2 + 0.1 = 0.5$

$$\frac{\partial L}{\partial w} = (\hat{y} - y) \cdot x = (0.5 - 1) \cdot 2 = -1$$

$$\frac{\partial L}{\partial b} = (\hat{y} - y) = (0.5 - 1) = -0.5$$

$w_{2new} \leftarrow 0.2 - 0.1 \cdot -1 = 0.3$

$b_{2new} \leftarrow 0.1 - 0.1 \cdot -0.5 = 0.15$

10. $s = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$ $\quad \text{softmax}(s) = \dfrac{e^s}{\sum_{n=1}^{3} e^s}$

$$\sum_{n=1}^{3} e^s = e^0 + e^1 + e^2 = 1 + e + e^2$$

$$\text{softmax}(0) = \frac{1}{1+e+e^2}$$

$$\text{softmax}(1) = \frac{e}{1+e+e^2}$$

$$\text{softmax}(2) = \frac{e^2}{1+e+e^2}$$

11.
Foward pass: takes input x apple w and b to get a prediction $\hat{y}$ utilizing an activation function while determining the loss based on the loss function. Backprop calculates gradient of the loss and utilizes it to adjust weights and biases.

12.
The network would just be a series of linear operations which would make it difficult for the model to learn non-linear patterns in data.

13.
Too large: changes are large so the model can make the loss function diverge → pointless.
Too small: changes to old w/b becomes miniscule and thus training becomes slow.

14.

Vanishing gradients : when gradients become extremely small due to multiplication over many layers. This prevents weights from updating meaningfully which makes training difficult.

15.

sigmoid : $\left(\frac{1}{1+e^{-z}}\right)' = \frac{e^{-z}}{(1+e^{-z})^2}$

if z is really large or small, the gradient is very nearly 0.

ReLU $(0, z)$ $\frac{\partial \text{ReLU}(z)}{\partial z} = 1$ when $z > 0$ and 0 when $z < 0$

prevents vanishing gradient.